

BEAMER: Behavioral Encoder to Generate Multiple Appropriate Facial Reactions

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ABSTRACT

This paper presents a framework for generating appropriate facial expressions for a listener engaged in a dyadic conversation. The ability to produce contextually suitable facial gestures in response to user interactions may enhance the user experience for avatars and social robots interaction. We propose a Transformer and Siamese architecture-based approach for generating appropriate facial expressions. Positive and negative Speaker-Listener pairs are created, applying a contrastive loss to facilitate learning. Furthermore, an ensemble of reconstruction quality sensitive loss functions is added to the network for learning discriminative features. The listener's facial reactions are represented with a combination of the 3D Morphable Model's coefficients and affect-related attributes (facial action units). The inputs to the network are pre-trained Transformer-based feature MARLIN and affect-related features. Experimental analysis demonstrate the effectiveness of the proposed method across various metrics in the form of an increase in performance compared to a variational auto-encoder-based baseline.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Behavioral Encoder, Contrastive Learning, Dyadic Interactions, Facial Reactions Generation, Transformer

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1 INTRODUCTION & BACKGROUND

Human communication is a fundamental aspect of our social structure, influencing our interactions, relationships, and overall well-being. Within the domain of interpersonal communication, dyadic human interactions characterized by one-to-one conversations between individuals hold a significant place [12]. It includes a wide variety of social interactions, ranging from informal discussions among friends to more formal exchanges in a professional setting. In a dyadic encounter, one person functions as the speaker and the other as the listener. The speaker conveys messages through verbal and nonverbal cues, while the listener replies through multiple means, such as facial expressions, voice, or bodily movements. This interactive process creates a dynamic exchange of information and emotions between the participants.

A key challenge in dyadic interactions lies in the uncertainty of the appropriate response to the speaker's behavior [12]. The interpretation and reaction to the speaker's cues can vary among different listeners or even within the same listener across different contexts. This variability in response can be influenced by many factors, including individual differences, cultural backgrounds, and personal experiences [9]. Exploration of these uncertainties and understanding the factors influencing the variability of responses in dyadic interactions can bring light to the complexities of human communication. As a result, more effective and context-sensitive human-computer interaction strategies can be designed. Typical use cases of automatic appropriate response generation are in user avatar interaction and social robotics. Figure 1 shows examples of multiple generated facial reactions for a single speaker by our developed model.

Although automatic generation of human-style reactions is an active area of research, in previous studies, researchers investigate the generation of human-style facial and bodily reactions where the main focus of these studies is on face generation and mesh reconstruction. Tesei *et al.* [15] use a Cyclic Generative Adversarial Network (GAN) architecture and propose image-to-image translation to transfer facial images from the original expression to different target expressions without paired examples. Deng *et al.* [5] propose a global and local adversarial networks-based approach for generating missing content in a UV map. This missing content is generated to be semantically consistent with the complete UV map obtained through 3D Morphable Model (3DMM) fitting on the 2D face images. This technique is valuable for preserving identity and generating visually realistic faces with diverse poses, contributing to advancements in facial generation techniques. These techniques

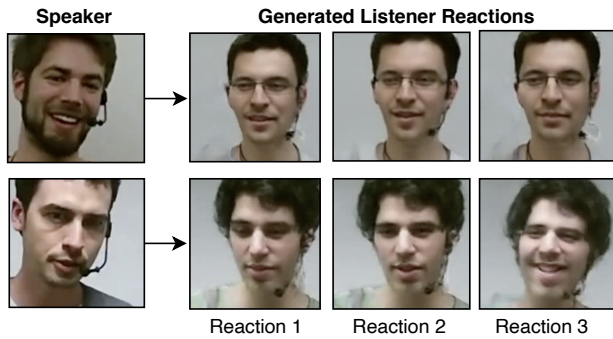


Figure 1: Multiple listener reactions generated via BEAMER. Column 1 are two frames from two speaker videos. Columns 2-4 are the generated reactions corresponding to the speaker face video for a listener.

primarily focus on generating predefined reactions that match the speaker’s actions, which makes generalization challenging as several non-verbal responses for similar speaker actions may exist.

Huang et al. [6] proposed a modified GAN architecture in which the generator is conditioned on the facial expressions of the conversation partner rather than the identity attributes of the generated person. Xu *et al.* [17] proposed a novel facial reaction generation framework which addresses the one-to-many facial reaction generation problem as a one-to-one mapping problem by summarizing the distribution of appropriate facial reactions displayed by listeners in response to each input speaker’s behavior.

This paper is a contribution to the ACM MM REACT Challenge 2023 [13], which aims to advance the automatic generation of multiple appropriate facial reactions (non-verbal) in response to a specific speaker’s behavior. The challenge comprises two sub-tasks: (a) offline multiple appropriate reactions generation (offline fMARG) and (b) online facial multiple appropriate reactions generation (online fMARG). In this paper, we focus on the former one, which involves developing a machine-learning model capable of accurately generating multiple realistic and natural spatio-temporal facial reactions based on a speaker’s behavior within a specified time period. The model input the entire sequence of a speaker’s behavior and generates multiple human-like facial reactions. These reactions contains appropriate Facial Action Units (AUs), facial expressions, and emotional states. To address the offline task in this paper, we propose a contrastive loss based behavioral encoder which learns appropriate listener and speaker non-verbal facial behavior into an embedding space representing appropriate listener reaction.

2 DATASET

We use three datasets: NoXI [2], RECOLA [10] and UDIVA [8]. NoXI is a collection of recorded interactions between individuals engaged in information retrieval tasks. It includes various languages, different topics, and unexpected situations. The RECOLA database consists of audio, visual, and physiological data from 46 French-speaking participants collaborating on a task. The UDIVA dataset contains video and audio data from pairs of participants engaged in collaborative and competitive tasks in English, Spanish, or Catalan. The audio-video data from these corpora are segmented into

30-second clips, and only dyadic interactions are included, where complete data of both conversational partners are present. After cleaning and preprocessing, the dataset comprises 8,616 clips, totaling approximately 71.8 hours of dyadic interactions [13].

2.1 Baseline Methods

The baseline [13] comprises of two methods: Transformer-VAE and BeLFusion [1]. The former method inputs a 30-second clip and generates a sequence of facial reactions of the same duration. The model is trained end-to-end with the mean square error loss to reconstruct the 2D facial frames. Additionally, a diversity loss and KL divergence loss are applied to increase sampling diversity and to constrain the predicted distribution tokens, respectively. The BeLFusion undergoes two stages of training. In the first stage, a VAE is trained to learn a condensed representation of visual features such as action units, facial affects, and expressions from a sequence of frames. Regression is integrated into the VAE, which transforms the reconstructed response into a sequence of 3DMM coefficients. The total VAE loss comprises KL divergence, reaction mean squared error, and 3DMM coefficients mean squared error. In the second stage, a latent diffusion model is trained to predict the lower-dimensional representation of the appropriate facial reaction of the listener based on the speaker’s reaction.

3 PROPOSED METHOD

We base our proposed architecture BEAMER (Behavioral Encoder to Generate Multiple Appropriate Facial Reactions) on a Transformer-VAE architecture [13] and create a behavioral encoder which takes into input a speaker’s behavior and texture and later encodes it into a latent space of an appropriate listener. The proposed method is depicted in Figure 2.

3.1 Feature Extraction

For encoding facial structure and texture, which contains useful clues regarding the perceived affect of an individual, we use the MARLIN features [3]. MARLIN is a pre-trained transformer, which has been learnt by focusing on densely masked facial regions like the eyes, nose, mouth, lips, and skin for reconstructing the complex spatio-temporal characteristics of a face video. This reconstruction process captures both local and global features, allowing for the encoding of general and transferable facial attributes. MARLIN is trained on web-crawled videos using self-supervised learning, enabling it to learn robust facial video features. By using MARLIN features, we reduce computational requirements compared to using raw video frames as features, while still obtaining rich, versatile, and reliable facial representations. These features consistently perform well across various downstream tasks. MARLIN’s emphasis on specific facial regions enables it to generate a meaningful representation of the face that incorporates fine-grained spatio-temporal details. This comprehensive representation can be advantageous for a wide range of applications and tasks.

We also extracted affect focused features, which include fifteen Facial Action Units (AUs) [7], two facial affect, and eight categorical facial expressions[13]. We combine MARLIN and affect features to build a strong feature space which describes the spatio-temporal facial movement.

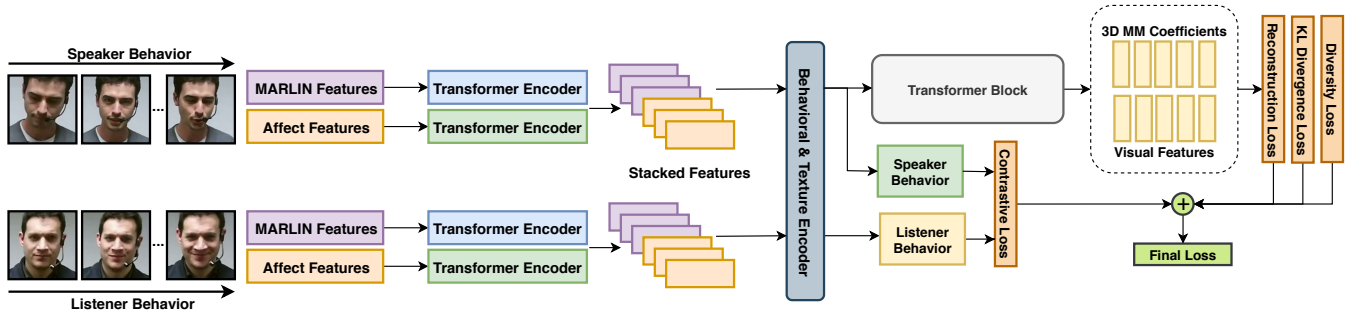


Figure 2: The proposed architecture for BEAMER. Pre-trained features and affect features are analysed through a transformer encoder followed by contrastive and reconstruction loss functions.

3.2 Behavioral and Texture Encoder

Speaker and listener videos are processed by extracting MARLIN¹ and affect features. These features are projected through a 1-D convolution layer followed by batch normalization, dense layers, and encoded by a transformer encoder independently. We introduce an encoder, which focuses on behavioral and texture information by first fusing these features sequentially and encodes them into a 128-dimensional latent vector for each speaker and listener.

3.3 Speaker-Listener Reaction Belongingness

Inspired by the recent works [4] in Siamese networks for evaluating belongingness (semantic similarity) of two signals, we add a contrastive loss between the speaker and the listener's features (Figure 2). The rationale is that the distance between the features of a speaker and an appropriate listener's reaction should be less as compared to an inappropriate listener's reaction.

We create negative speaker-listener pairs by randomly selecting a listener's reaction from the same event in the same dataset. For example, if a speaker is from the NoXI dataset, then the negative listener pair will be chosen from the NoXI dataset under the same event/context. We use cosine similarity as a loss metric to compute contrastive loss between the positive and negative speaker-listener pairs which are appended to the overall loss of the system.

3.4 Transformer Block

Transformers leverage self-attention mechanisms, allowing them to focus on different parts of the input sequence. [11, 16] This attention mechanism enables us to effectively capture both local and global dependencies, which becomes beneficial for understanding complex relationships while handling latent feature representations.

The speaker's latent representation is processed by a transformer encoder to predict the parameters of the latent distribution. We reuse the Transformer Decoder architecture provided in their baseline. [13]. We train BEAMER with 4 loss functions - MSE for the reconstruction of the visual features and 3D coefficients, KL-Divergence for the distance between the ground truth and generated data distributions, diversity loss to maintain diversity for each generated sample from the same speaker's behavior, and contrastive loss between speaker and listener's behavior. The weight of KL-Divergence and diversity loss functions were set to 0.0002 and 10,

respectively inspired by [13]. The rest of the loss functions have equal weightage. We train BEAMER by flipping speaker and listener pairs as a data augmentation strategy. For example, if one instance is [speaker_1, listener_1], then the flipped instance becomes [listener_1, speaker_1]. Since we aren't using audio modality and in the context of a dyadic conversation, this strategy becomes relevant and becomes an appropriate speaker-listener behavior.

3.5 Evaluation Metrics

The evaluation metrics are based on four aspects: appropriateness, diversities, synchrony, and realism. [14] The appropriateness is evaluated by estimating the similarity between generated and their corresponding real facial reactions. Two metrics, *FRD* (Dynamic Time Warping) and *FRC* (Concordance Correlation Coefficient) are calculated, where *FRD* measures the distance between real and generated reactions, and *FRC* measures their correlation. Diversities refer to the range of variations observed within a specific frame and between different frames. In order to quantify these diversities, metrics like *FRVar*, *FRDiv*, and *FRDvs* are applied. Synchrony refers to the assessment of the harmony between generated facial expressions and the corresponding behavior of the speaker. *FRSyn* (Time Lagged Cross Correlation) is applied to measure synchrony.

4 RESULTS

We compare our approach to the baseline Transformer-VAE and BeLFusion method and present the results in terms of appropriateness, diversity, and synchrony metrics. The overall scores on the *Validation* and *Test* sets are provided in Table 1 and Table 2, respectively. Also, we show two examples of generated facial reactions, given a single speaker's behavior in Figure 1. This Figure shows 3 generated facial reactions.

4.1 Appropriateness

Appropriateness of a predicted facial reaction refers to how well it resembles the ground truth listener's reaction and the expected set of appropriate facial reactions in the same context. It is measured by two metrics - *FRD* and *FRC*. *FRD* quantifies the similarity of the predicted facial reaction with respect to the closest appropriate facial reaction. *FRC* measures how closely the predicted listener's facial reaction resembles the ground truth listener's facial reaction.

¹<https://github.com/ControlNet/MARLIN>

Table 1: Results on the Validation set.

Method	Appropriateness		Diversity			Synchrony
	FRC(↑)	FRD(↓)	FRDiv(↑)	FRVar(↑)	FRDvs(↑)	FRSyn(↓)
Trans-VAE	0.12	101.37	0.0265	0.0045	0.0322	49.00
BeLFusion (k=1)	0.12	94.39	0.0087	0.0058	0.0106	47.14
BeLFusion (k=10)	0.14	94.26	0.0134	0.0078	0.0149	46.94
BeLFusion (k=10) + Binarized AUs	0.12	97.17	0.0323	0.0173	0.0341	49.00
BEAMER	0.07	96.20	0.0361	0.0084	0.0589	49.00

Table 2: Results on the Test set.

Method	Appropriateness		Diversity			Synchrony
	FRC(↑)	FRD(↓)	FRDiv(↑)	FRVar(↑)	FRDvs(↑)	FRSyn(↓)
Trans-VAE	0.10	98.48	0.0242	0.0040	0.0263	44.88
BeLFusion (k=1)	0.12	90.21	0.0085	0.0056	0.0103	44.95
BeLFusion (k=10)	0.13	89.84	0.0137	0.0078	0.0149	45.02
BeLFusion (k=10) + Binarized AUs	0.12	92.58	0.0322	0.0170	0.0337	49.00
BEAMER	0.07	95.40	0.0240	0.0046	0.0420	49.00

The results in Table 1 and Table 2 achieved by BEAMER surpass the baseline metrics when compared with Trans-VAE in terms of distance between the prediction and real appropriate facial reactions (*FRD*). BEAMER outperforms by 5.17 points in *FRD* metric for the obtained results on the *Validation* dataset. For the *Test* set, there is a performance improvement of 3.08 points corresponding to *FRD*. BEAMER did not perform well on the *FRC* metric, which indicates a lack of correlation between the generated 3DMM coefficients and affect features when compared with the ground truth for a few samples. Overall, we observed that appropriateness metrics increased by 5.22 points and 3.11 points on the *Validation* and *Test* sets respectively as compared to [13].

4.2 Diversity

FRDiv, *FRVar*, *FRDvs* metrics serve as measurement tools to assess the extent of diversities BEAMER is able to generate. This is an important metric to identify the variations associated with a specific speaker-listener in a frame-by-frame setting. The intra-subject diversity metric is defined by *FRDiv* where BEAMER outperforms the baselines on the *Validation* set and achieves comparable performances on the *Test* set for baselines except for BeLFusion with binarized AUs.

The results obtained for *FRDiv* on the *Validation* set by BEAMER outperform the Trans-VAE baseline and for the *Test* set the outcomes are comparable. *FRVar* measures the intra-sequence diversity. BEAMER performs better by 0.0039 points on the *Validation* set as compared with the baseline. For the *Test* set, *FRVar* metric outcome is almost equivalent. BEAMER shows improvement in terms of inter-subject diversity (*FRDvs*) on *Validation* and *Test* sets. with 0.0267 points and 0.0157 points, respectively over the baseline. Further, an improvement of 0.0402 points on the diversity metric for the *Validation* set is observed. On *Test* set, there is an absolute increase of 0.0161 for the diversity attribute.

4.3 Synchrony

The *FRSyn* metric is applied to evaluate synchrony in the data points and it computes TLCC between input speaker behavior and generated listener reactions. We observe a similar score for BEAMER and baseline Trans-VAE for the findings on the *Validation* set. Whereas on the *Test* set, the results for BEAMER are higher by 4.12 points. Noticeably, there is a decrease in performance for the synchrony metric by 4.12 points.

5 CONCLUSION & FUTURE WORK

We present BEAMER a behavioral and texture encoder which aims to align the speaker’s behavior features with the listener’s appropriate reaction in the latent space. The contrastive approach between speaker and listener features is effective in learning discriminative features for the network, which generates appropriate listener reaction. Further, we found that the combination of MARLIN and affect related features are useful for the generation task. Our proposed approach outperforms the Trans-VAE baseline in 4 out of 6 evaluation metrics. Additionally, it outperforms the BeLFusion baseline specifically for the *FRDiv* and *FRDvs* metrics. Furthermore, we consistently observe that the proposed approach achieves superior performance in the *FRDvs* metric on both the *Validation* and *Test* sets. This observation validates that the proposed approach has the capability to generate diverse appropriate listener reactions.

While our primary focus in this work is on Visual modality, we suggest exploring the Audio modality in future research to improve overall appropriateness, diversity, and synchrony. Furthermore, we observe a limitation of our approach when evaluating the *FRC* metric which identifies multiple appropriate listener behaviors for each speaker’s behavior. We would like to experiment with loss functions sensitive to the correlation with ground truth and the generated reactions. Furthermore, data augmentation strategies can be applied to augment the joint latent representations of the speaker-listener embeddings, which essentially bring robustness and improve generalization capabilities.

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